

Deleted File Recovery using Python

Python Program:

```
import os

with open("secret_note.txt", "w") as f:
    f.write("MEETING_POINT: warehouse 7, 11pm")

with open("secret_note.txt", "rb") as f:
    original_bytes=f.read()

os.remove("secret_note.txt")
print("File 'secret_note.txt' deleted.")
print("Exists on disk now?", os.path.exists("secret_note.txt"))

disk_buffer=original_bytes+b"\x00"*50
marker=b"MEETING_POINT"
index=disk_buffer.find(marker)

if index!=-1:
    recovered=disk_buffer[index:].split(b"\x00")[0]
    with open("recovered_note.txt", "wb") as f:
        f.write(recovered)
    print("Recovered content:", recovered.decode())
else:
    print("No recoverable data found.")
```

Sample Output:

```
File 'secret_note.txt' deleted.
Exists on disk now? False
Recovered content: MEETING_POINT: warehouse 7, 11pm
```

Explanation:

This program demonstrates a simple deleted file recovery simulation. It creates a text file, reads and stores its contents, deletes the file, and then simulates scanning raw disk data for a known byte pattern. When the marker is found, the program reconstructs and saves the deleted information into a new file. This illustrates the basic concept of deleted file recovery used in digital forensics.